# Shuo Wu

*Ph.D. Candidate, ECE | LLM Systems: Retrieval & Ranking, Routing & Memory, Evaluation*

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## Summary

Ph.D. candidate (expected **2027**) building LLM systems that decide what to compute under a budget: which documents to read (**retrieval, RAG and reranking**: **AV-IP**, sole-author zero-shot LLM reranker, eight BEIR datasets), which model to call (**routing, cascades and strategy memory**: **Inter-Cascade**, co-first author, TMLR 2026, up to **48%** fewer strong-model calls and **50%** lower API cost), and which expert to query (online learning and bandits, **UAI 2026**). Builds the PyTorch and Hugging Face **evaluation harnesses** and **confidence-calibration** guarantees behind every result. Eight papers and preprints: **TMLR 2026** (co-first author), first-author **JAIR 2025**, UAI 2026, ISIT 2025; ICLR 2027 under review.

## Education

**University of Illinois Chicago**, Chicago, IL — Sep 2022–2027 (expected)
**Ph.D., Electrical and Computer Engineering**; Advisors: Prof. Shuo Han and Prof. Lev Reyzin

**Southern University of Science and Technology**, Shenzhen, China — Sep 2018–Jul 2022
**B.S., Mathematics and Applied Mathematics**

## Research Experience

**Inter-Cascade: In-Context Knowledge Distillation for LLM Cascades** — Oct 2024–Aug 2025
*Co-first author; TMLR 2026; arXiv:2509.22984*

- Co-developed **Inter-Cascade**, an online LLM cascade that fixes a core inefficiency of static cascades (re-escalating repeated or similar queries to the expensive model): on each deferred query the strong model writes a reusable **problem-solving strategy**, stored in a dynamic repository.
- Cut inference cost (strong-model calls by up to **48.05%**, API cost by up to **49.63%**) while raising weak-model accuracy by up to **33.06%** (overall system by up to 6.35%).
- Strategies are retrieved by **FAISS** similarity search and injected into the weak model's context, so it improves at inference time through **in-context learning** with no fine-tuning. Proved the paper's theorem that this augmentation improves the weak model's **confidence calibration**. Works with open-source and API LLMs.
- Engineered the end-to-end evaluation stack: batched inference with response caching, automated ablations, and error slicing by segment.

**AV-IP: Zero-Shot LLM Reranking Below the Softmax** — Mar 2025–Present
*Sole author; under review, ICLR 2027*

- Proposed **AV-IP**, a **zero-shot LLM reranker** that scores documents by value-vector agreement instead of attention mass: on Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct across **eight BEIR datasets**, the worst AV-IP variant beats the best whole-network attention-mass baseline on macro **nDCG@10**.
- Unified prior attention-based zero-shot rerankers (ICR, Selective-ICR, QRHead, ReAttn, HeadRank) under **one formula**; an ablation (attention weights and value vectors held fixed) and a substitution control isolate the query's value direction as the ranking signal.
- Proved that value centering replaces the calibration forward pass, so AV-IP needs **no second pass and no document KV cache** across passes; that pass costs mass-based baselines 30–49% extra **latency** (or 3.7–4.2 nDCG points if dropped), about zero for AV-IP.

**Skill Extraction from Noisy Text: NLP Pipeline and Evaluation Harness** May 2026–Present
*Segmentation and multi-level evaluation for labor-market skill analysis*

- Built a **multi-level evaluation harness** (segment-, document-, and union-coverage metrics) and a hybrid sentence-segmentation pipeline for noisy federal job postings (spaCy dependency splitting with a neural Segment-any-Text fallback).
- Recovered **10.1%** more unique skills per document and **2.2x** more fully covered documents than the JAAT toolkit's NLTK segmenter.

**Data Filtering for Model-Based Reinforcement Learning** Jul 2024–Oct 2024
*Co-author; arXiv:2410.12160*

- Integrated an **out-of-distribution rollout filter** (drops model rollouts that diverge from real data) into **MBPO** and built the experiment pipeline (sweeps, logging, MuJoCo evaluation).
- Matched MBPO returns with about **20%** fewer environment interactions on MuJoCo continuous control, without a model ensemble.

## Technical Skills

**LLM systems:** Hugging Face Transformers, LLM APIs (open-source & API models), in-context learning & prompting, model **cascades, routing & deferral** (multi-model orchestration), **strategy memory**, knowledge distillation, response caching, KV cache, inference cost & latency
**Retrieval, search & NLP:** natural language processing, RAG, embeddings & vector search (FAISS), BM25, reranking, zero-shot ranking, BEIR, nDCG@10, sentence segmentation (spaCy)
**Evaluation:** LLM evaluation harnesses, benchmark pipelines, ablations & substitution controls, error slicing, confidence calibration, deferral policies, coverage metrics
**RL & decision-making:** model-based RL (MBPO, MuJoCo), MDPs, reward design, bandits, online learning, budgeted expert & annotator selection
**Machine learning, deep learning & tools:** Python, PyTorch, NumPy, scikit-learn, Gurobi, ILP/MILP, Git/GitHub, Linux, pytest, GitHub Actions, LaTeX

## Selected Publications & Preprints

* = equal contribution. Papers with L. Reyzin use alphabetical author order (theory convention).

- **Attention is the Address, Values are the Message: Zero-Shot Reranking Below the Softmax.** *Under review, ICLR 2027.*
  Shuo Wu.
- **From Deferral to Learning: Online In-Context Knowledge Distillation for LLM Cascades.** *Transactions on Machine Learning Research (TMLR)*, 2026. arXiv:2509.22984.
  Yu Wu*, Shuo Wu*, Ye Tao*, Yansong Li, Anand D. Sarwate.
- **Margin-Based Boosting for Partial Interpretability.** *Manuscript, 2026.*
  Xing Gao, Lev Reyzin, Shuo Wu.
- **Expert Advice with Costly Observations.** *Conference on Uncertainty in Artificial Intelligence (UAI), 2026.*
  Lev Reyzin, Aadirupa Saha, Shuo Wu.
- **Margin-Based Sparsification for Feature-Budgeted Linear SVM.** *International Symposium on Artificial Intelligence and Mathematics (ISAIM), 2026.*
  Lev Reyzin, Shuo Wu.
- **Robust Reward Design for Markov Decision Processes.** *Journal of Artificial Intelligence Research (JAIR), Vol. 84, 2025.*
  Shuo Wu, Haoxiang Ma, Jie Fu, Shuo Han.
- **Observational Learning with a Budget.** *IEEE International Symposium on Information Theory (ISIT), 2025.*
  Shuo Wu, Pawan Poojary, Randall Berry.
- **When to Trust Your Data: Enhancing Dyna-Style Model-Based Reinforcement Learning With Data Filter.** arXiv:2410.12160 (2024).
  Yansong Li*, Zeyu Dong*, Ertai Luo, Yu Wu, Shuo Wu, Shuo Han.